

Cystoscopy in Dogs and Cats



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KEYWORDS

- Cystoscopy • Urethroscopy • Urinary incontinence • Ectopic ureter • Hematuria
- Pollakiuria • Urinary obstruction • Urinary cancer

KEY POINTS

- For experienced, well trained clinicians, cystoscopy is a minimally invasive tool that has advanced urology in veterinary medicine.
- Cystoscopy provides the ability to diagnose conditions that cannot easily be diagnosed using other modalities.
- Cystoscopy enables thorough and minimally invasive assessment and treatment of conditions diagnosed using other modalities.
- As technology advances, a wider range of patients will be able to be assessed via cystoscopy, and more cystoscopic guided treatment options will become available.
- Advanced training and a wide range of specialized equipment are necessary to perform cystoscopy appropriately.

INTRODUCTION: NATURE OF THE PROBLEM

Cystourethroscopy is a technique used to gain access to the lower genitourinary tract (urethra, urinary bladder, ureteral orifices, vagina). In most cases, cystourethroscopy is used as a diagnostic tool to visually assess the lower urinary tract if routine diagnostic evaluation (blood work, urinalysis, urine culture, radiography, ultrasonography) does not yield a definitive diagnosis for the cause of a patient's lower urinary tract disease. In addition, some treatment modalities can be administered with cystoscopic guidance. Cystourethroscopy is considered a minimally invasive procedure but does require general anesthesia in order to minimize patient movement and secondary iatrogenic injury to the lower urinary tract.

INDICATIONS FOR CYSTOURETHROSCOPY

Persistent/Recurrent Lower Urinary Tract Signs

Common lower urinary tract signs in dogs and cats include dysuria, pollakiuria, stranguria, hematuria, and inappropriate urination (**Boxes 1** and **2**). The most common

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differentials for these signs in dogs are urinary tract infection (**Fig. 1**), urolithiasis (**Fig. 2**), and lower urinary tract neoplasia (**Fig. 3**), whereas in cats feline idiopathic cystitis (also known as feline lower urinary tract disease, **Fig. 4**) is also a common differential for lower urinary tract signs.¹ Cystoscopy is indicated when routine diagnostic evaluation (blood work, urinalysis, urine culture, radiography, ultrasonography) does not yield an obvious cause for a patient's lower urinary tract signs. It is also indicated when there has been an apparent therapeutic failure for a diagnosed cause of lower urinary tract signs. An example of this is a patient with lower urinary tract signs that do not abate after appropriate antibiotic therapy is instituted for a urinary tract

Box 1

Indications for cystoscopy

Assessment of persistent or recurrent lower urinary tract signs of unknown origin

- Urethral or bladder masses
- Urethral strictures
- Occult uroliths

Assessment of persistent or recurrent urinary tract infections

- Investigate for anatomic abnormalities that may predispose the patient to development of infections
- Identify urethral or bladder masses
- Identify uroliths
- Identify vestibulovaginal stenosis
- Identify and resect (via laser) vestibulovaginal septal remnants

Assessment of chronic hematuria

- Identify bladder or urethral masses
- Identify occult uroliths
- Identify primary renal hematuria (hematuria seen exiting ureteral orifices)

Assessment of bladder or urethral masses

- Assess extent of mass
- Cystoscopic guided biopsy
- Cystoscopic guided laser resection of cystic polyps

Treatment of cystic or urethral calculi

- Cystoscopic guided stone basketing
- Laser lithotripsy
- Electrohydraulic lithotripsy
- Laparoscopic assisted cystoscopic urolith removal
- Percutaneous cystolithotomy

Assessment and treatment of urinary incontinence

- Assess for the presence of ectopic ureters
- Cystoscopic guided laser ablation of ectopic ureters
- Cystoscopic guided submucosal injection of urethral bulking agents

Box 2	
Cystoscopic guided treatment options for various urinary tract disease processes	
Disease Processes Diagnosed via Cystoscopy and Treatable with Cystoscopic Guided Procedures	
Disease Process	Cystoscopic Guided Treatment Options
Cystic or urethral calculi	1. Retrograde cystoscopic guided stone basketing 2. Retrograde cystoscopic guided laser lithotripsy 3. Antegrade percutaneous cystolithotomy
Cystic polypoid masses	Cystoscopic guided laser resection
Idiopathic renal hematuria	Cystoscopic guided sclerotherapy
Ureteral ectopia (intramural)	Cystoscopic guided laser ablation of ectopic ureters
Urethral sphincter mechanism incompetence (presumptively diagnosed when other causes of urinary incontinence are ruled out)	1. Cystoscopic guided submucosal bulking agent injections 2. Cystoscopic guided botulinum toxin injections

infection diagnosed via urine culture and sensitivity testing. It is possible that such a patient may have a concurrent condition, such as occult urolithiasis or urinary tract neoplasia. Similarly, a patient that has previously undergone a cystotomy for urolith extraction but continues to have lower urinary tract signs may have suture remnants present in the urinary bladder (Fig. 5).²

Evaluation of Hematuria

Hematuria can be caused by bleeding from any portion of the urinary or genital tract. Cystoscopy is a useful tool for the diagnosis of the source and cause of hematuria in dogs and cats. It is particularly useful in patients for which thorough diagnostic

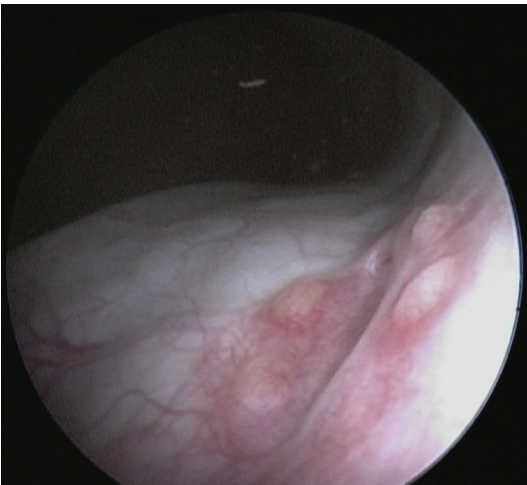


Fig. 1. Cystoscopic image from a 2-year-old female spayed (FS) terrier cross dog with urinary tract infection and urinary bladder wall pustules.

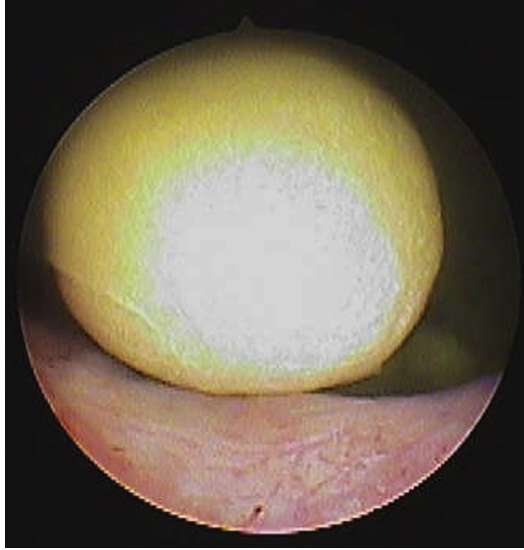


Fig. 2. Cystoscopic image from a 7-year-old female spayed bichon frise with struvite and calcium phosphate carbonate uroliths.

evaluation, including coagulation testing, does not diagnose a definitive cause of hematuria. In patients that do not have obvious inflammatory lesions (**Fig. 6**) within the lower urinary tract, primary renal hematuria (**Fig. 7**) becomes a likely differential. This condition can be diagnosed definitively by monitoring the urine jets (**Fig. 8**) from each ureteral orifice for obvious bleeding.^{3,4} Cystoscopic guided sclerotherapy can be performed in select patients with primary renal hematuria.



Fig. 3. Cystoscopic image from a 12-year-old female spayed mixed breed dog with a transitional cell carcinoma at the trigone.

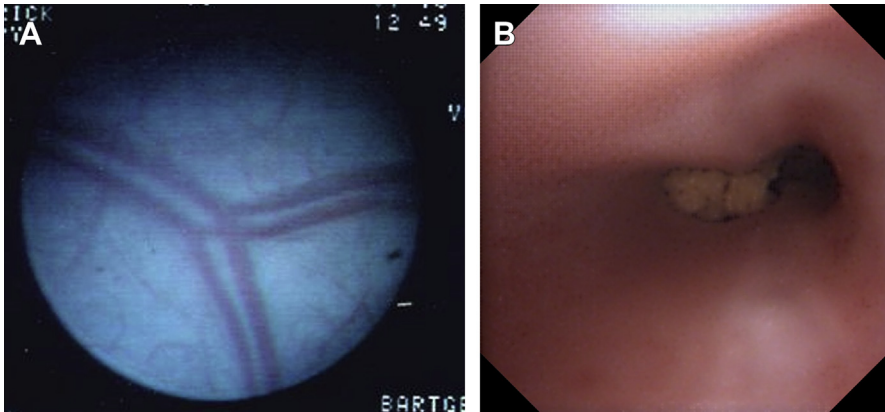


Fig. 4. (A) Cystoscopic image from a 3-year-old spayed female domestic short hair (DSH) cat with vascular congestion secondary to feline lower urinary tract disease. (B) Cystoscopic image from a male dog of unknown age with a urethrolith at the base of the os penis.

Urinary Tract Masses

Cystoscopy can be helpful in the diagnosis of lower urinary tract masses. In many cases, these masses are diagnosed using other modalities (most commonly transabdominal ultrasonography). Occasionally, cystoscopy reveals masses not visualized using other diagnostic tools, particularly when they are located only in the urethra, vagina, or vestibular vault. Transitional cell carcinoma and prostatic carcinoma are the most common malignant neoplasms of the lower urinary tract, but other types of neoplasia (rhabdomyosarcoma, leiomyosarcoma, lymphosarcoma, hemangiosarcoma, fibrosarcoma, metastatic neoplasia) have also been reported.^{5–20} Cystoscopic guided mass biopsy can be performed via introduction of a flexible biopsy instrument

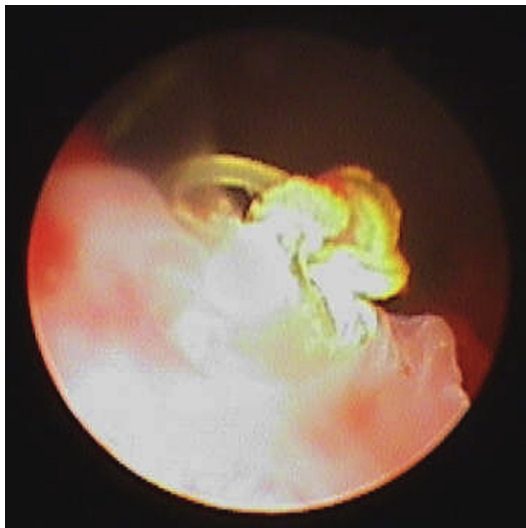


Fig. 5. Cystoscopic image from a 9-year-old female spayed Lhasa apso with a suture knot in the urinary bladder.



Fig. 6. Cystoscopic image from a 5-year-old female spayed Labrador dog with hemorrhagic cystitis secondary to recurrent UTIs.

through the cystoscopy sheath's instrument port.^{21–23} Inflammatory polyps are common benign masses of the urinary bladder. Inflammatory polyps can appear similar to transitional cell carcinoma based on visual inspection ([Fig. 9](#)), and cystoscopic guided biopsy must be performed to definitively differentiate the two. Inflammatory polyps often have a stalklike base ([Fig. 10](#)), and are most commonly located at the apex of

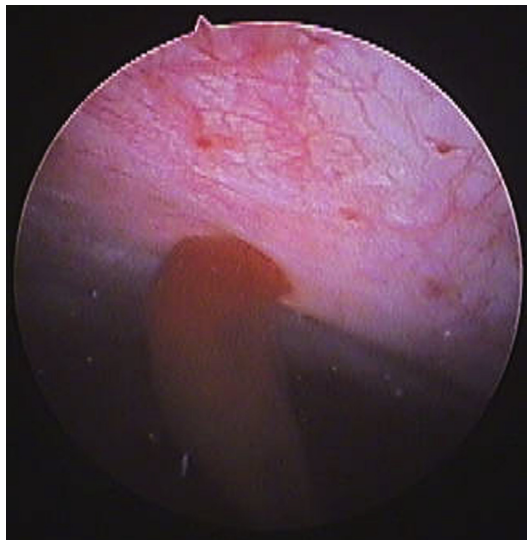


Fig. 7. Cystoscopic image from a 5-year-old female spayed mixed breed dog with bloody urine passing from the right ureteral orifice secondary to primary renal hematuria.

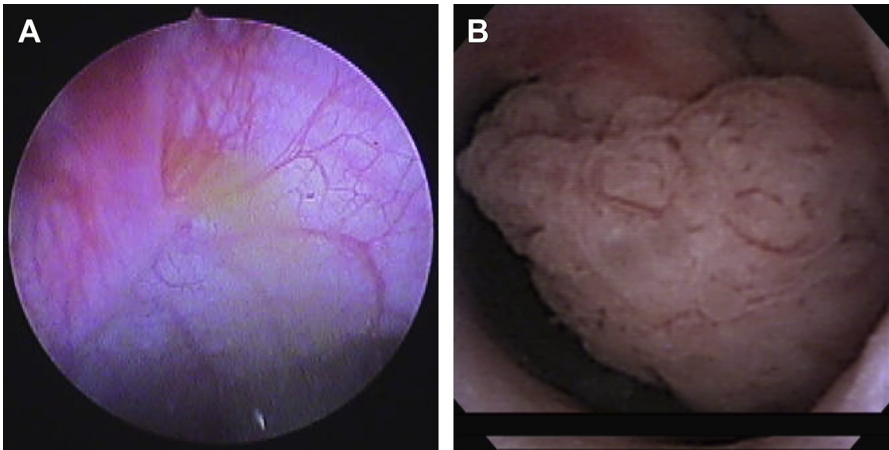


Fig. 8. (A) Cystoscopic image from a 7-year-old female spayed Yorkshire terrier dog with a normal left ureter passing yellow urine. (B) Cystoscopic image from a male dog of unknown age with urinary bladder transitional cell carcinoma.

the urinary bladder, whereas transitional cell carcinoma masses are most commonly located at the bladder trigone.^{24,25} Holmium yttrium aluminum garnet lasers or diode lasers can be used to ablate the stalks of inflammatory polyps with cystoscopic guidance.²⁶ A cystoscopic stone basket or cystoscopic biopsy instrument can then be used to grasp the polyp and remove it from the urinary tract through the urethra. Care must be taken to ensure that the polyp is of an appropriate size to be removed through the urethra, because an inability to remove the polyp after detachment could result in a risk of urinary obstruction. Proliferative urethritis (also known as granulomatous urethritis) is a severe, proliferative, inflammatory process that occurs in the urethra and which, on gross examination, can appear identical (Fig. 11) to urethral neoplasia.²⁷

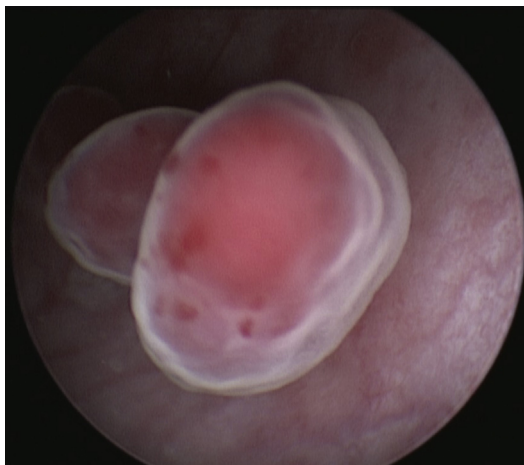


Fig. 9. Cystoscopic image from a male dog of unknown age with polypoid cystitis secondary to stones.

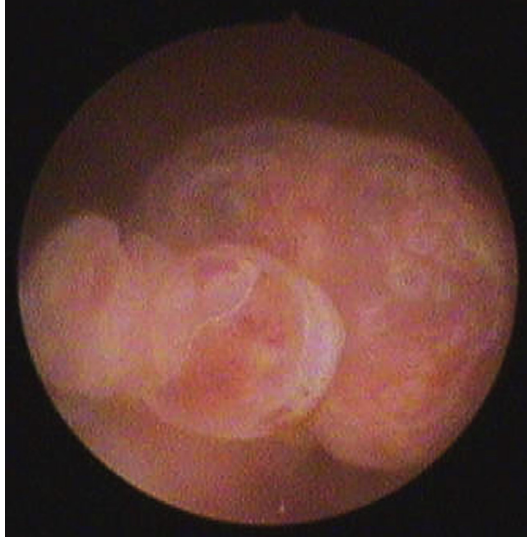


Fig. 10. Cystoscopic image from a 10-year-old female spayed Lhasa apso with polypoid cystitis.

Urolithiasis

Cystoscopy can be helpful in the diagnosis of uroliths not visualized with other common diagnostic tools, such as radiolucent calculi that are located in areas not able to be visualized by ultrasonography (eg, distal urethra). In patients previously diagnosed with urolithiasis via other modalities, cystoscopy can be used to gain access to the calculi ([Fig. 12](#)) and then cystoscopic assisted removal can be performed. For small

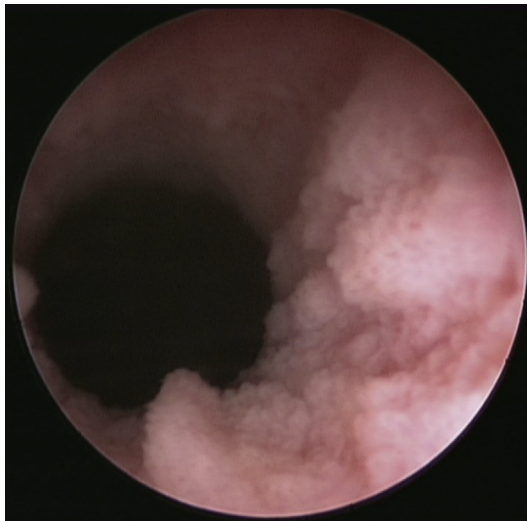


Fig. 11. Cystoscopic image from a 9-year-old female spayed retriever cross dog with lymphoplasmacytic urethritis.

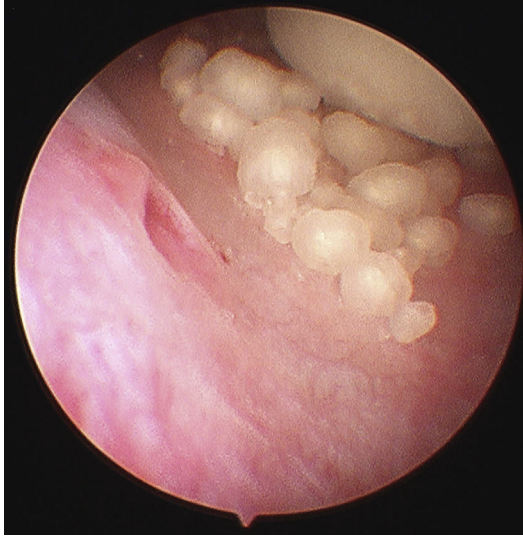


Fig. 12. Cystoscopic image from a female spayed dog of unknown age with uroliths in the urinary bladder. Note the normal right ureteral orifice.

calculi that can pass easily through the urethra, a stone basket (**Figs. 13 and 14**) can be passed through the cystoscope instrument port and used to grasp the calculi and remove them from the urinary bladder. Previous literature on voiding urohydropulsion provides information about the largest uroliths that were able to pass through the urethras of patients of varying sizes, sexes, and species (**Box 3**).²⁸ This information can be used to help determine the likelihood that cystoscopic guided stone basketing would be a viable option for urolith removal in a patient. For larger calculi, a lithotripsy fiber

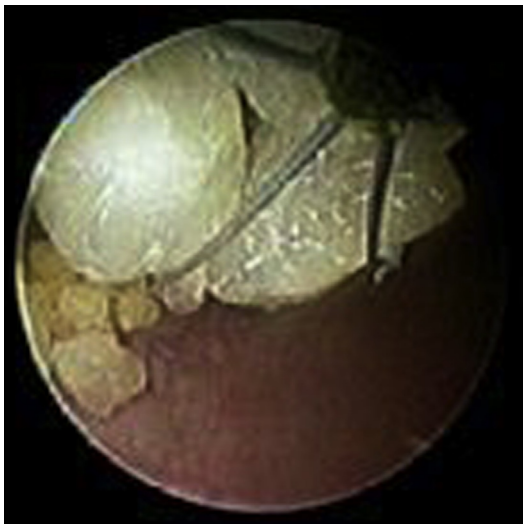


Fig. 13. Cystoscopic image from a female spayed dog of unknown age having multiple uroliths retrieved by a stone basket forceps.

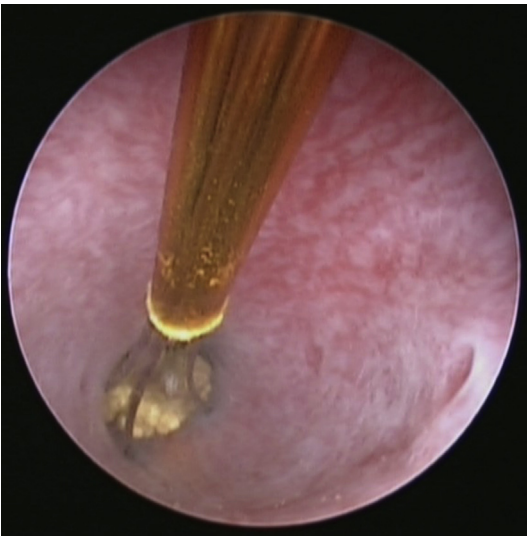


Fig. 14. Cystoscopic image from a female spayed dog of unknown age having urethrolith retrieved by graspers.

(Fig. 15) can be passed through the cystoscope instrument port and used to fragment the calculi into smaller pieces that can then be removed through the urethra via stone basketing or voiding urohydropulsion.^{29–32} Because of the small size of the urethras of male dogs, and the resultant need to use a flexible cystoscope (Figs. 16 and 17), laser lithotripsy procedure times can be prolonged and there is a higher risk of incomplete urolith removal in male dogs than in female dogs.^{29,31} For these reasons, antegrade approaches to the lower urinary tract have been described as a minimally invasive alternative to laser lithotripsy or cystotomy in male dogs.^{33–35} An antegrade approach may also be preferable to laser lithotripsy for female cats, particularly those with a large stone burden.³⁶ This approach also provides a minimally invasive option for stone removal in male cats.

Recurrent Urinary Tract Infections

Cystoscopic evaluation of the lower urinary tract can reveal anatomic abnormalities that may predispose the patient to recurrence of UTIs. Examples of such abnormalities

Box 3			
Maximum diameter of uroliths passed through the urethra via voiding urohydropulsion			
Maximum Urolith Size Passed During Voiding Urohydropulsion			
Species	Sex	Body Weight (kg)	Maximum Urolith Size (mm)
Canine	Female	7.4	7
	Male	9	5
Feline	Female	4.6	5
	Male	6.6	1

From Lulich JP, Osborne CA. Voiding urohydropulsion: a nonsurgical technique for removal of urocystoliths. In: Bonagura JD, editor. Kirk’s current veterinary therapy small animal practice. 12th edition. Philadelphia: WB Saunders; 1995:1006; with permission.

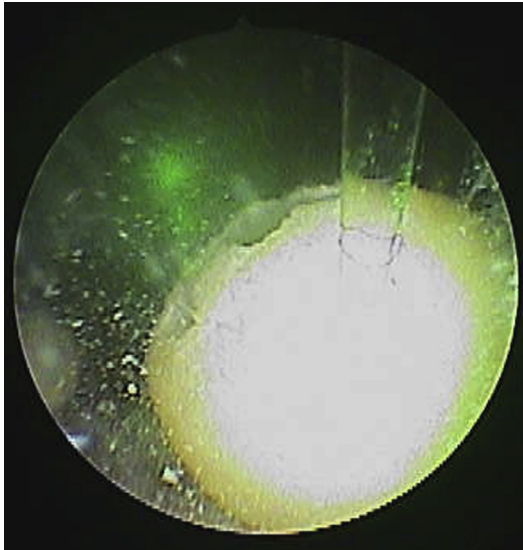


Fig. 15. Cystoscopic image from a 7-year-old female spayed bichon frise with struvite and calcium phosphate carbonate uroliths managed with lithotripsy.

include ectopic ureters (**Figs. 18 and 19**), urinary masses (**Figs. 20 and 21**), uroliths, vestibulovaginal stenosis, urachal diverticulum (**Fig. 22**), persistent paramesonephric remnant (hymen), vaginourethral fistula, and urethrorectal fistula.^{37–42} In some cases, these abnormalities can be corrected cystoscopically (intramural ectopic ureters, uroliths, inflammatory polyps, persistent paramesonephric remnants).^{29,31,37,43–46}

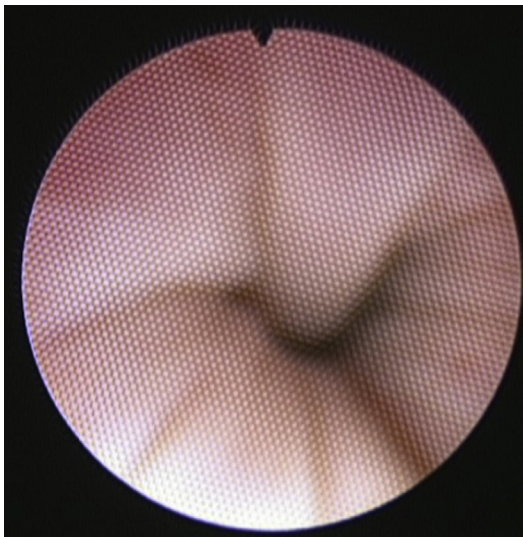


Fig. 16. Cystoscopic image from a male dog of unknown age with normal urethra. The image was obtained from a fiberoptic cystoscope.

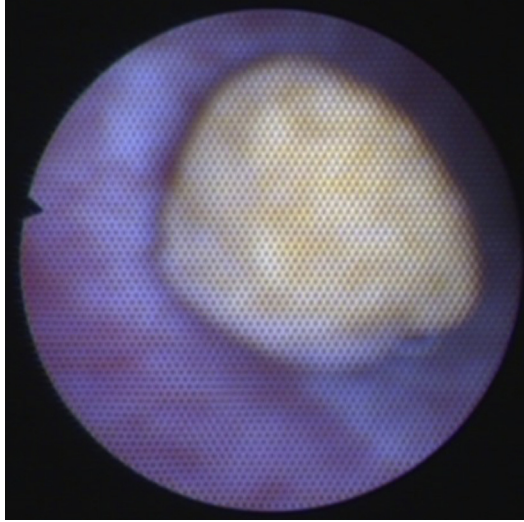


Fig. 17. Cystoscopic image from a male dog of unknown age with a urethrolith.

Urinary Incontinence

Cystoscopic evaluation of the lower urinary tract can reveal anatomic abnormalities resulting in urinary incontinence. Examples include ectopic ureters, intrapelvic bladder (short urethra), urethral hypoplasia (**Fig. 23**), vaginourethral fistula, and persistent vestibulovaginal septae.^{37,38,45–48} Intramural ectopic ureters can be ablated via cystoscopic laser ablation.^{43,44,46} Patients with no obvious anatomic or neurologic causes for urinary incontinence are typically presumed to have urethral sphincter

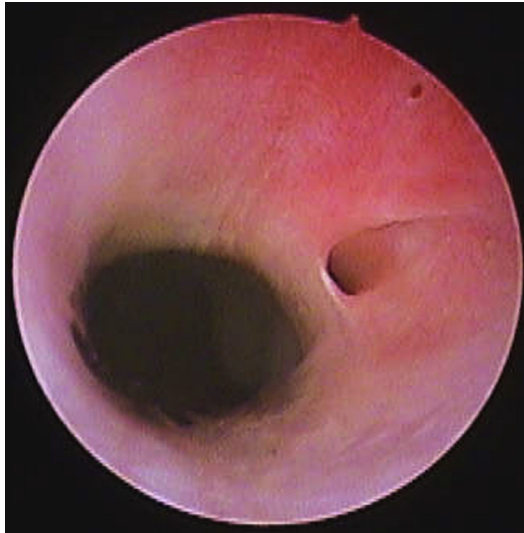


Fig. 18. Cystoscopic image from a 4-year-old female spayed Labrador dog with ectopic ureter (tear-shaped opening on the right).

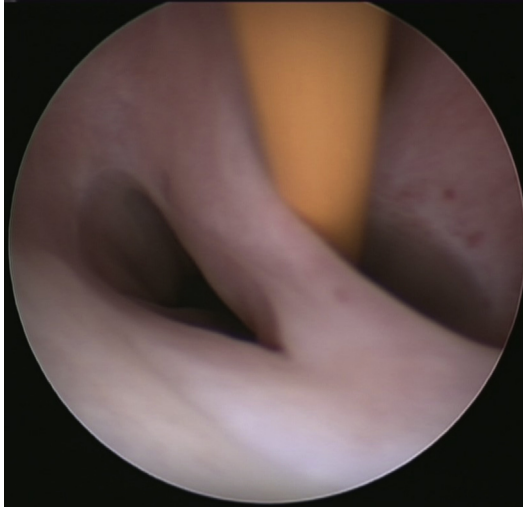


Fig. 19. Cystoscopic image from an 8-month-old female spayed rottweiler with an ectopic ureter (catheter in ectopic ureter).

mechanism incompetence. Urodynamics, including urethral pressure profilometry and cystometrography can be performed in patients with refractory incontinence to confirm abnormalities in urethral and/or urinary bladder tone, but have limited availability. Patients with urethral sphincter mechanism incompetence can be treated with cystoscopic guided submucosal bulking agent injections ([Fig. 24](#)) or possibly botulinum toxin injections.^{49–52}

CONTRAINDICATIONS

Thrombocytopenia or Other Coagulopathy

Cystoscopy is minimally invasive and induces minimal trauma to the urinary tract when performed correctly ([Box 4](#)). However, in patients with defective primary or secondary



Fig. 20. Cystoscopic image from a male dog of unknown age with polypoid cystitis.

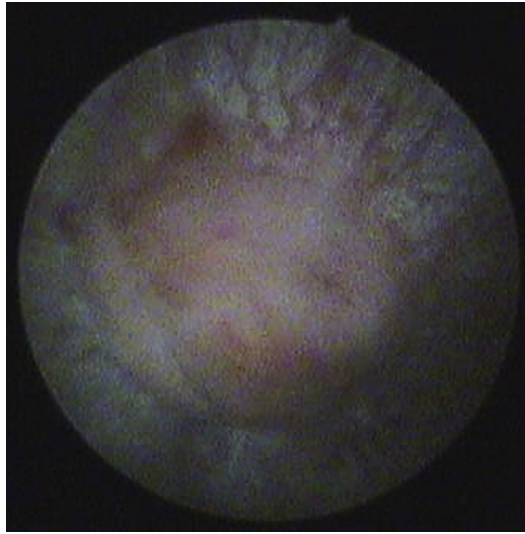


Fig. 21. Cystoscopic image from a 10-year-old female spayed miniature poodle with a sterile suture-induced urinary bladder granuloma.

hemostasis, even mild trauma could result in significant hemorrhage, potentially placing the patient at risk of blood loss as well as the formation of obstructive urethral blood clots. In addition, intraoperative hemorrhage can cause deterioration of the cystoscopic image quality, thus decreasing the diagnostic utility of the procedure.

Anesthetic Intolerance

In dogs and cats, cystoscopy should be performed under general anesthesia because of the risk of iatrogenic injury to the urinary tract if the patient is excessively mobile during the procedure. Alterations to the anesthetic protocol can be made based on an individual patient's needs.

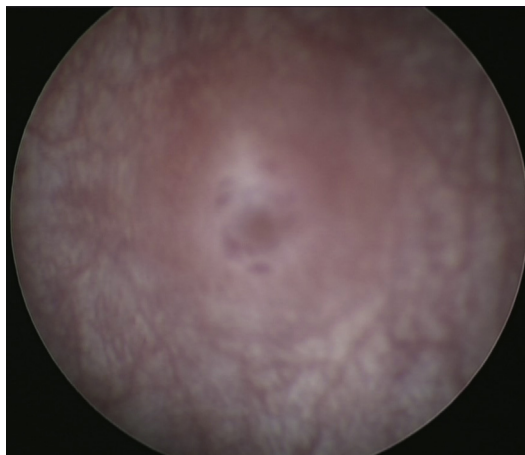


Fig. 22. Cystoscopic image from a male dog of unknown age with a urachal diverticulum.



Fig. 23. Cystoscopic image from a 9-month-old female intact (FI) rottweiler with urethral hypoplasia.

Ruptured Urinary Bladder or Urethra, Recent (<7 Days) Urinary Tract Surgery

Cystoscopy on patients with rupture of any portion of the urinary tract can be challenging because of a loss of normal anatomic markers and difficulty distending the bladder or urethral lumen.^{22,26} In addition, the vestibular vault, vagina, and distal urethra are not sterile environments, leading to a risk of introducing bacteria to the abdominal cavity (bladder and proximal urethral rupture) or the perineal region (distal urethral rupture).⁵³

Complete Urethral Obstruction

Obstruction of the urethra could be considered a contraindication to cystourethroscopy in that complete urinary obstruction may render thorough assessment of the

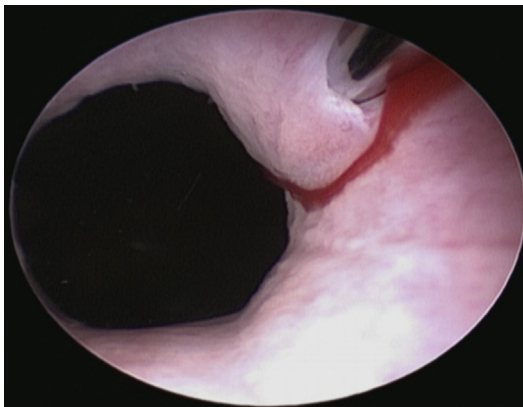


Fig. 24. Cystoscopic image from a 6-year-old female spayed boxer with urinary incontinence treated with urethral bulking polydimethylsiloxane.

Box 4**Contraindications to cystoscopy**

Coagulopathy
 Anesthetic intolerance
 Bladder/urethral rupture
 Recent cystotomy or urethrotomy (<7 days before procedure)

entire lower urinary tract impossible. However, urethroscopy can be used to determine the cause of the obstruction and can potentially be part of the treatment of the problem (eg, laser lithotripsy of urethroliths).²⁶

TECHNIQUE**Equipment**

A variety of cystoscopes is needed to perform transurethral cystoscopy in both male and female dogs and cats because of anatomic variations and marked variability in urethral diameter. A rigid cystoscope can be used in female dogs and cats. Flexible cystoscopes are required in male dogs and cats for complete evaluation of the urethra and urinary bladder. Most rigid cystoscopes are composed of 2 parts (**Fig. 25**): the inner telescope, which permits the conduction of light and the image; and the outer sheath, which protects the telescope and has ports and channels for the instillation of sterile liquid (or gas); drainage of urine; and passage of biopsy forceps, stone removal baskets, or lithotripsy fibers. The sheath also provides a smooth edge to protect the urethral and bladder mucosa. The sheath has a locking device to connect it to the telescope. Some cystoscopes have been developed that contain both the sheath and telescope in 1 unit. This design provides a slightly smaller outer diameter.^{2,26} Telescopes are available in a wide range of sizes and lengths. The most commonly used cystoscope is a 2.7-mm diameter, 18-cm-long cystoscope with a 14.5-Fr sheath. This cystoscope is appropriate for female dogs greater than 5 kg and less than 20 kg. In dogs larger than 15 to 20 kg, the 2.7-mm cystoscope typically permits visualization of the urethra and bladder trigone, but not a complete bladder examination. In these larger female dogs, a 3.5-mm diameter, 30-cm-long cystoscope with a 17-Fr sheath provides the needed length. In female dogs smaller than 5 kg and female cats, a 1.9-mm diameter, 18-cm-long cystoscope with a 10-Fr sheath is required. Image quality is typically better with the larger telescopes. Telescopes are available with a

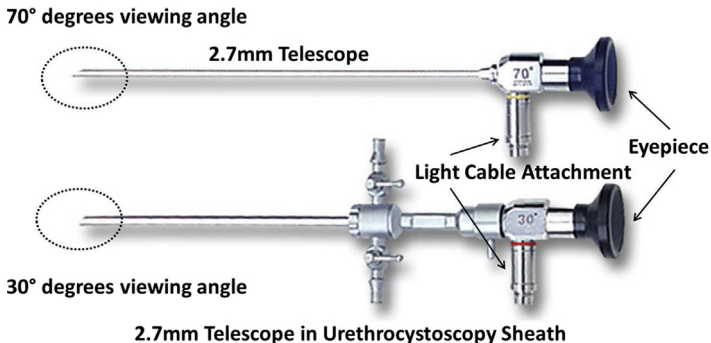


Fig. 25. Labeled 2.7-mm telescope cystoscope with and without a sheath.

variety of viewing angles, but a 30° angle of view is most commonly used in veterinary medicine.^{2,26} Cystoscopy in male dogs requires a flexible cystoscope because of the anatomy of the urethra. These endoscopes are classified as ureteroscopes or ureterorenoscopes by most companies that sell cystoscopic equipment, because these endoscopes are used to visualize the ureters and renal pelvis in humans. Multiple flexible cystoscopes are available, with the most commonly used endoscope being 7.5 Fr and 55 to 100 cm long (**Fig. 26**). When selecting a flexible cystoscope it is important to consider the degree of deflection of the distal tip (ideal is 270° dual deflection), particularly if interventional procedures are to be performed. Cystoscopy in male cats is limited by the small luminal diameter of the male cat urethra but is possible with a flexible 1.2-mm diameter or semirigid 1.0-mm cystourethroscope. Both endoscopes provide visualization only. They have no biopsy channels and have poor image and light transmission quality because of the limited numbers of fiber optic bundles. Alternative techniques that could be used to image the urinary bladder and proximal urethra in male cats are rigid retrograde cystourethroscopy using a 1.9-mm, 18-cm-long rigid cystoscope following perineal urethrostomy, or antegrade cystourethroscopy after obtaining access to the urinary bladder surgically or laparoscopically.^{2,26,34,35} Additional equipment needed to perform cystoscopy includes a light source (xenon light sources provide the best light quality), light-transmitting cable, sterile infusion fluid (typically 0.9% NaCl [saline] or lactated Ringer solution [LRS]), and irrigation lines. A suction device is also needed for flexible cystoscopy. Many endoscopists also use a high-definition camera (with connector) that attaches to the eyepiece of the telescope and transfers the image to an image processor. The image processor projects the image on a video monitor. This equipment can be stored on an endoscopy tower. A separate camera is also frequently used for flexible cystoscopy performed with a fiber optic endoscope but is not needed with a video endoscope. An image capturing device permits storage of images and/or videos for documentation in the patient medical record, future comparisons, and client education. Biopsy forceps, stone removal baskets or nets, electrocautery, or lithotripsy instruments are needed if these interventions are to be performed. After being cleaned and air dried, rigid telescopes and flexible cystoscopes should be stored in storage bins designed to avoid compression damage.^{2,26}

Preparation

Cystoscopy should be performed under general anesthesia.

Female dogs or cats

The perivulvar region is clipped and cleaned with surgical scrub. An antiseptic vaginal flush (0.05% chlorhexidine acetate solution [12.5 mL of chlorhexidine added to 500 mL



Fig. 26. Flexible cystoscope.

of sterile LRS or a 0.9% NaCl bag makes a 0.05% solution]) should be used before cystoscopy, particularly in patients with excessive vaginal discharge.

Male dogs

Preputial and peripreputial hair is clipped and the preputial and peripreputial region is cleaned with surgical scrub. The prepuce is then flushed with 0.05% chlorhexidine acetate solution for 2 minutes.⁵⁴

Male cats

Preputial and peripreputial hair is clipped and the preputial and peripreputial region is cleaned with surgical scrub. Stay sutures are placed through the reflection of the prepuce at the base of the penis in order to facilitate penile extrusion.²²

Patient Positioning

Female dogs or cats

The authors prefer to perform rigid cystoscopy in dorsal recumbency because of an improved ability to access the ureteral orifices for procedures such as ureteral stent placement or cystoscopic guided laser ablation of ectopic ureters, and decreased chance of fecal contamination of the vulva in dorsal recumbency.⁵⁵ Lateral recumbency and sternal recumbency have also been described.^{2,56} If dorsal recumbency is used, the patient is placed in a radiology positioning trough located at the end of the examination table. The tail base is positioned at the edge of the trough. The hind limbs are tied in a frog-legged position. A sterile patient drape is placed over the patient and a hole is made to expose the vulva.

Male dogs

Lateral recumbency is used most commonly, but male dogs may also be positioned in dorsal recumbency, which offers some benefits with regard to complete visualization of the ureteral orifices.⁵⁶ A sterile patient drape is placed over the patient, and a hole is made to expose the prepuce. Care is taken to ensure that the cranial aspect of the drape is sufficiently long to reduce the risk of contamination of the flexible cystourethroscope.

Male cats

The patient is placed in dorsal recumbency. A patient drape is placed over the patient, and a hole is made to expose the prepuce.

Approach and Procedure

Female dogs: rigid retrograde cystourethroscopy

The rigid cystoscope should be fully assembled (telescope, sheath, camera, light source, light transmitting cable, irrigation lines connected) before introduction into the lower urinary tract. The rigid cystoscope should be liberally lubricated with sterile lubricating jelly before passage into the vestibular vault. The vulva is grasped and gently retracted caudally using the thumb and index finger of the endoscopist's nondominant hand. The cystoscope is then passed using the dominant hand between the fingers that are retracting the vulva, thus creating a seal around the cystoscope with the vulvar folds. Care is taken not to introduce the cystoscope into the clitoral fossa (Fig. 27). Irrigation is introduced to allow filling and complete assessment of the anatomy of the vestibular vault. In dorsal recumbency, the vaginal vault is located at the bottom of the image, and the urethral os is located at the top of the image (Fig. 28). The cystoscope is not initially advanced into the vagina, which has a larger number of bacteria than the urethra.⁵³ The cystoscope is slowly passed into the



Fig. 27. A female spayed dog of unknown age with a normal clitoral fossa.

urethral os and irrigation is used to distend the urethra in order to enable visualization before cystoscope advancement. In order to prevent iatrogenic urethral trauma, care must be taken to keep the lumen of the urethra toward the bottom of the field of view because of the 30° angle of the rigid cystourethroscope. The cystoscope is gently advanced through the urethra and into the urinary bladder. If resistance is met at any point during cystoscope advancement through the urethra, the endoscopist stops advancing the cystoscope and withdraws the cystoscope slightly to visualize the

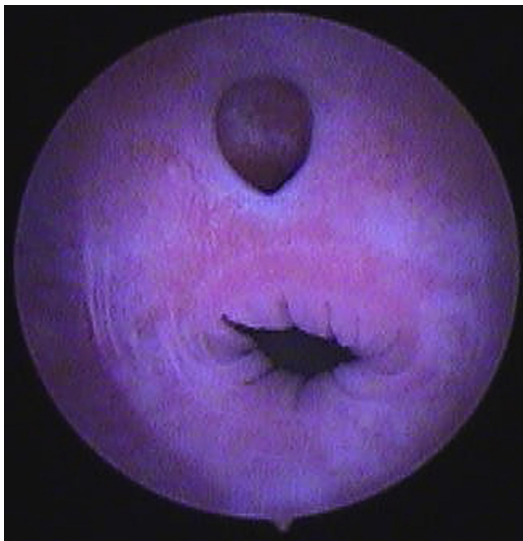


Fig. 28. Cystoscopic image from a 10-year-old female spayed miniature poodle with a normal vaginal vestibule distended with fluid (the vagina is the lower opening; the external urethral opening is at the top).

urethra and determine the reason for resistance. If there is no obvious reason for resistance, then it is most likely that the endoscopist is not appropriately centering the cystoscope in the lumen of the urethra (ie, the lumen is not being kept at the bottom of the screen) and therefore the cystoscope tip is encountering the dorsal wall of the urethra. The best way to correct this problem is for the endoscopist to lower the wrist slightly, such that the urethral lumen drops to the bottom of the screen. Once the cystoscope enters the urinary bladder, it is centered within the bladder lumen and the egress port is opened to allow the urinary bladder to be emptied. A urine sample should be obtained for analysis and bacterial culture if not already performed. Once the bladder is empty, it is filled with sterile saline. If the view of the bladder wall is still cloudy because of the presence of turbid urine, the bladder is again emptied of fluid and refilled with saline until the bladder wall is able to be visualized clearly. The entire bladder wall is carefully assessed (Fig. 29). The cystoscope sheath is moved counter-clockwise while the camera is held stable such that the light source connection to the cystoscope is pointing upward. By doing this, the 30° angle of the cystoscope tip is pointing downward, which enables excellent visualization of both ureteral orifices. The ureteral orifices are C-shaped structures that face each other on the dorsal wall of the urinary bladder's trigone. The orifices are assessed with regard to their positioning within the urinary bladder, their anatomy, and the production of a jet of normal urine from each ureteral orifice (see Fig. 6). The cystoscope is then gradually withdrawn from the urethra. Following cystourethroscopy, vaginoscopy (see Fig. 29A) should be performed to assess for vestibulovaginal septal remnants (see Fig. 29B, C), vaginal masses (see Fig. 29D, E), vaginitis (see Fig. 29F), or vaginourethral fistulae.

Female cats: rigid retrograde cystourethroscopy

The rigid cystourethroscope should be fully assembled (telescope, sheath, camera, light source, light transmitting cable, irrigation lines connected) before introduction into the lower urinary tract. The cystoscope should be liberally lubricated with sterile lubricating jelly before passage into the vestibular vault. Saline irrigation is started before inserting the cystoscope into the vulva so that the appropriate rate of irrigation can be determined. Aggressive irrigation can result in overdistension of the urinary bladder and potential bladder rupture. The vulva is gently grasped and retracted caudally using the thumb and index finger of the endoscopist's nondominant hand. The cystoscope is then passed using the dominant hand between the fingers that are retracting the vulva, thus creating a seal around the cystoscope with the vulvar folds. The cystoscope is pointed downward so that the vestibular vault can be accessed. The urethral opening appears as a vertical slit at the top of the field of view, whereas the vaginal opening is a small circular opening at the bottom of the field of view (Fig. 30).^{21,22,36} The cystoscope is gently passed into the urethral orifice and is gently advanced through the urethra, which is longer and narrower in female cats than in female dogs.^{21,22,36} The urethra of female cats is often most narrow at its distal aspect, so entering the urethral orifice can be challenging in some female cats.³⁶ Care must be taken to keep the lumen of the urethra toward the bottom of the field of view because of the 30° angle of the rigid cystourethroscope in order to prevent iatrogenic urethral trauma. The cystoscope is gently advanced through the urethra and into the urinary bladder. Starting in the midurethra, there is a smooth ridge of tissue that arises from the dorsal wall of the urethra. This ridge is present along the remaining length of the urethra, up to the urethrovesicular junction. If resistance is met at any point during cystoscope advancement through the urethra, the endoscopist stops advancing the cystoscope and withdraws the cystoscope slightly to visualize the urethra and determine the reason for resistance. If there is no obvious reason for

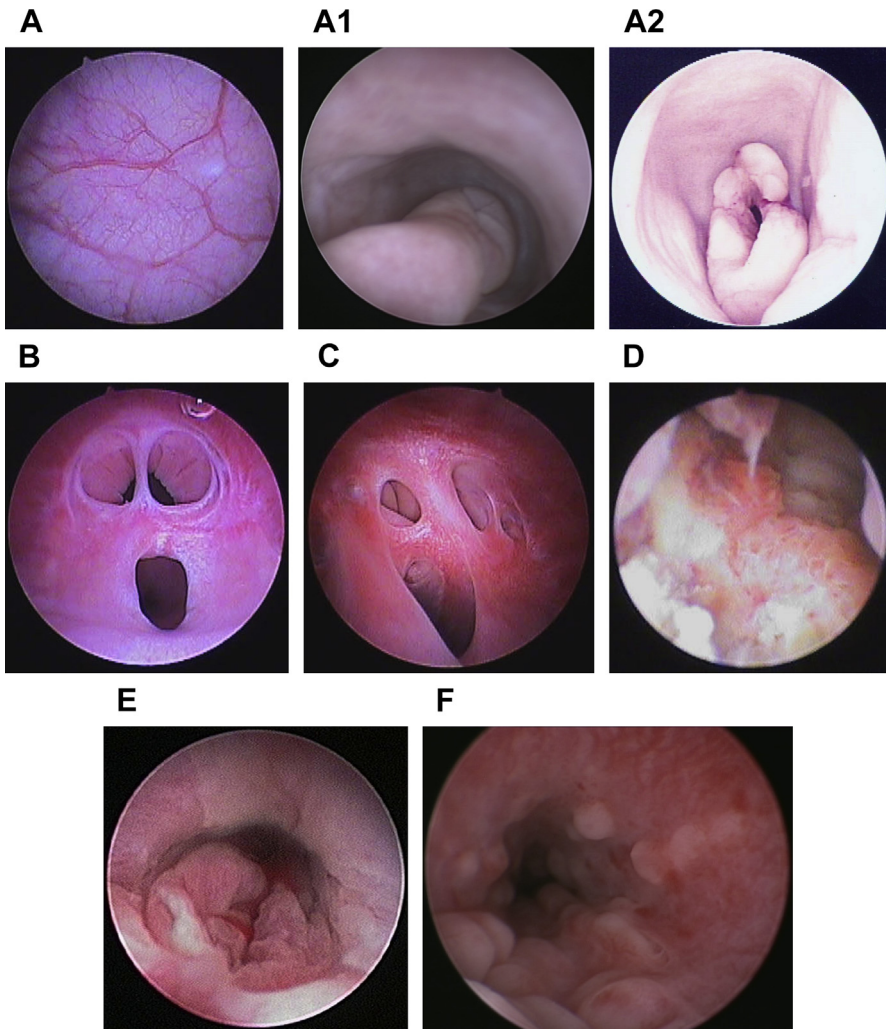


Fig. 29. (A) Cystoscopic image from a 7-year-old female spayed Yorkshire terrier with normal urinary bladder mucosa. (A1) Vaginoscopy image from a female spayed anestrous dog of unknown age with close cervical os. (A2) Vaginoscopy image from a 4-year-old female intact (FI) beagle with open cervical os in estrous. (B) A 7-year-old female spayed Yorkshire terrier with a thin frenulum in the vaginal vestibule. (C) Cystoscopic image from a 4-year-old Labrador with complex wide band frenulum (the patient also had bilateral ectopic ureters, which are not visible in this image). (D) A 9-year-old female spayed Norwegian elkhound with vaginal squamous cell carcinoma mass extending to the level of the external urethral papilla. (E) Vaginoscopy from a female spayed dog of unknown age with a vaginal transitional cell carcinoma. (F) Vaginoscopy from a female intact dog of unknown age with vaginitis (the patient also had urethritis, which is not visible in this image).

resistance, one possibility is that the endoscopist is not appropriately centering the cystoscope in the lumen of the urethra (ie, the lumen is not being kept at the bottom of the screen), and therefore the cystoscope tip is encountering the dorsal wall of the urethra. As is the case in female dogs, the best way to correct this problem is for the

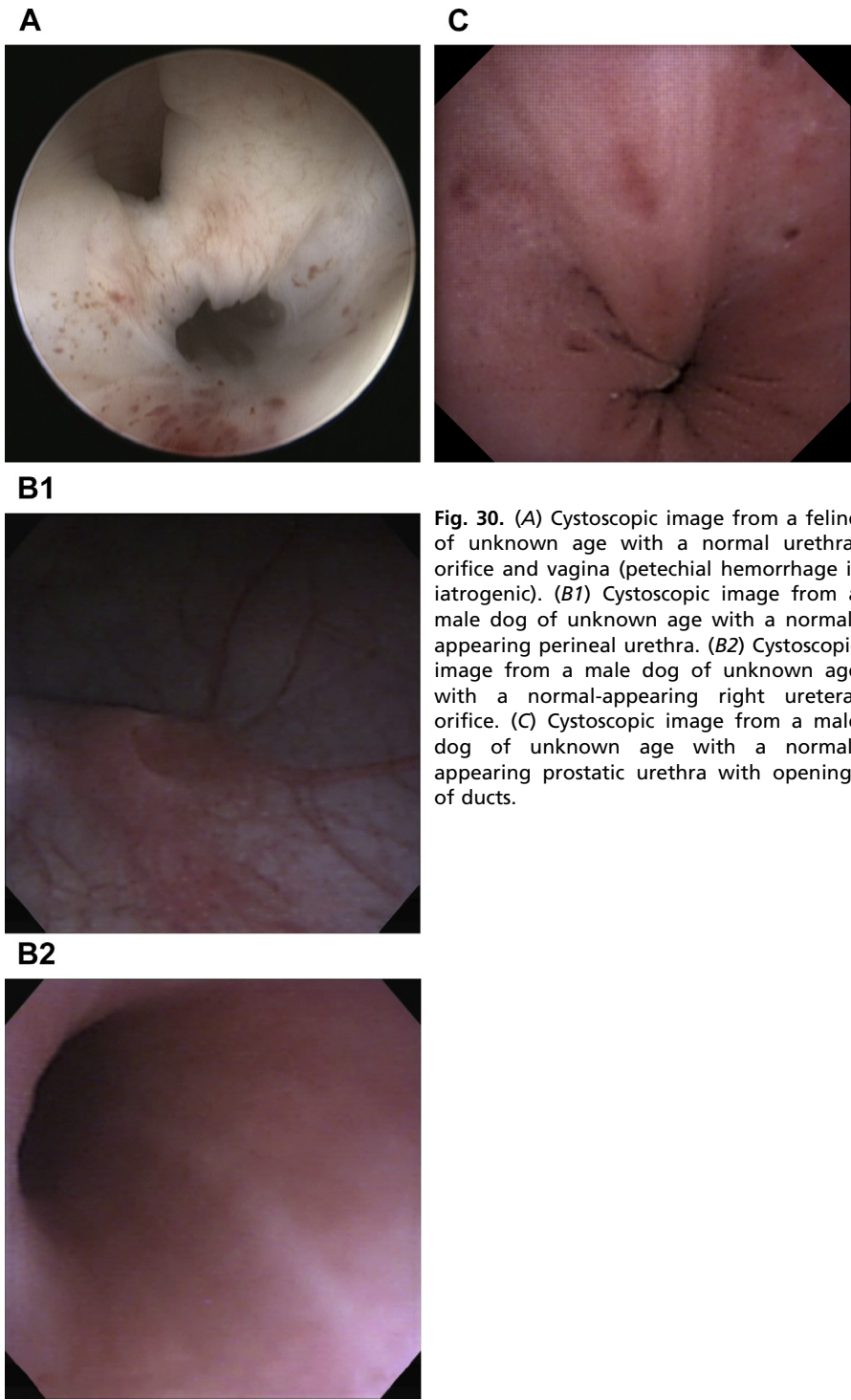


Fig. 30. (A) Cystoscopic image from a feline of unknown age with a normal urethral orifice and vagina (petechial hemorrhage is iatrogenic). (B1) Cystoscopic image from a male dog of unknown age with a normal-appearing perineal urethra. (B2) Cystoscopic image from a male dog of unknown age with a normal-appearing right ureteral orifice. (C) Cystoscopic image from a male dog of unknown age with a normal-appearing prostatic urethra with openings of ducts.

operator to lower the wrist slightly, such that the urethral lumen drops to the bottom of the screen. If resistance is still met, then it is most likely that the urethra is too narrow for cystoscopy. Retrograde cystourethroscopy should be aborted, and antegrade cystourethroscopy should be performed instead.³⁶ Once the cystoscope enters the urinary bladder, it is centered within the bladder lumen and the egress port is opened to allow the urinary bladder to be emptied. A urine sample should be obtained for analysis and bacterial culture at this time if not already performed. Once the bladder is empty, it is filled with saline. If the view of the bladder wall is still cloudy because of the presence of turbid urine, the bladder is again emptied of fluid and refilled with saline until the bladder wall is able to be visualized clearly. The entire bladder wall is carefully assessed. The cystoscope sheath is then moved counterclockwise while the camera is held stable such that the light source connection to the cystoscope is pointing upward. By doing this, the 30° angle of the cystoscope tip is pointing downward, which enables visualization of both ureteral orifices. Unlike female dogs, the ureteral orifices of female cats are located in the most proximal urethra.^{22,36} The ureteral papillae arise from the previously described ridge of tissue on the dorsal urethral wall, and each ureteral orifice appears as a small slit on the top of each ureteral papilla.³⁶ The orifices are assessed with regard to their positioning, their anatomy, and the production of a jet of normal urine from each ureteral orifice (**Fig. 31**). The cystoscope is then gradually withdrawn from the urethra.

Male dogs: flexible retrograde cystourethroscopy

The authors typically empty the urinary bladder using a red rubber catheter attached to a 60-mL syringe before the procedure. The penis is extruded from the prepuce by an assistant. The penis is then gently grasped and the flexible cystourethroscope is passed into the urethral orifice. Care is taken not to enter the prepuce instead of the urethra, because the prepuce is a nonsterile environment. An assistant facilitates the infusion of saline through the cystoscope into the urethra. Because the working channel of flexible cystourethroscopes is narrow, using only a hanging bag of saline for infusion does not typically provide an adequate stream of saline to distend the urethra. The authors typically use a system in which the

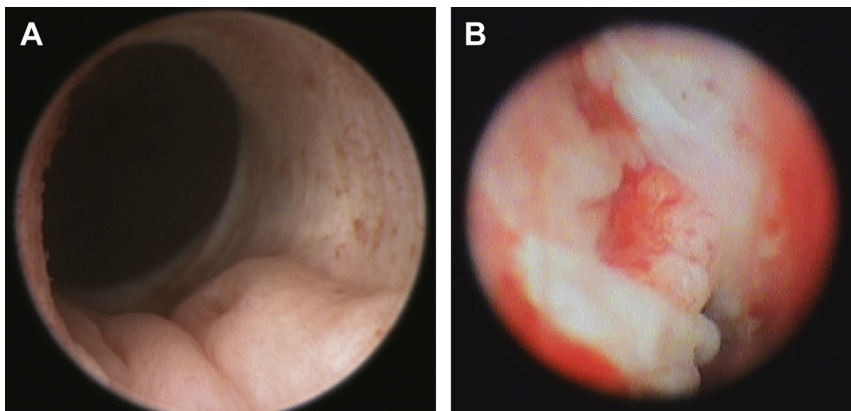


Fig. 31. (A) Cystoscopic image from a feline of unknown age with normal ureterovesicular junctions at the urinary bladder neck. (B) Cystoscopic image from an 8-year-old female spayed Labrador with small distal urethral perforation (noted at the 5 o'clock position) and urethral transitional cell carcinoma.

primary infusion line connected to the hanging bag of saline is attached to a 3-way stopcock. The other 2 ports of the stopcock are connected to an extension line that enters the ingress/egress port of the flexible cystourethroscope, and a 60-mL syringe. Saline is drawn into the 60-mL syringe from the saline bag, and is then infused into the cystoscope. Using this system, a higher pressure stream of saline can be used to dilate the urethral lumen, which drastically improves visualization of the urethra. Unlike female dog and cat cystourethroscopy, the urethral lumen should be kept in the center of the image as the cystourethroscope is advanced through the urethra, because the flexible cystourethroscope has a 0° angle tip. Once the prostatic urethra is entered, the prostatic ductules can be seen opening into the urethra. These ductules should not be mistaken for ectopic ureteral orifices. The bladder is then entered. The bladder is mostly empty and saline is infused to distend the bladder's lumen and the bladder's wall is thoroughly assessed. One major downside of flexible cystourethroscopy is that complete assessment of the bladder wall can be challenging because of the limitations of using a fiber optic scope with a narrow field of view. As the cystourethroscope is backed out of the urinary bladder, the cystoscope tip is deflected dorsally so that the ureteral orifices can be visualized. Assessment of the ureteral orifices using flexible cystourethroscopy can be challenging, and the authors typically wait to see a jet of urine in order to guide the cystourethroscope to the appropriate location to assess the orifices. Following assessment of the urinary bladder and ureteral orifices, the cystourethroscope is gradually backed out of the urethra and the urethra is assessed a second time. Following assessment of the urethra and urinary bladder, the preputial fossa is examined with the cystourethroscope.

Male cats: semiflexible retrograde urethroscopy

This technique can only be used to assess the urethra of male cats.²² If evaluation of the urinary bladder is necessary, antegrade cystourethroscopy is recommended (discussed later). Two operators are required for this procedure. The penis is kept extruded from the prepuce by preplaced stay sutures in the reflection of the prepuce. The penis is gently grasped and the semirigid cystoscope is passed into the urethral orifice. One operator (assistant) controls the infusion of saline through the injection port. The second operator (endoscopist) controls advancement of the cystourethroscope. The urethral lumen must be kept in the center of the image. Because the tip of the semirigid cystourethroscope cannot be deflected, this depends on tension exerted on the penis, saline infusion into the cystourethroscope, and torque placed on the cystourethroscope.²² The normal bulbocavernosus glands and prostate appear as narrowings in the urethra. The entire urethra is carefully assessed as the cystourethroscope is advanced and withdrawn from the urethra.

Antegrade cystourethroscopy

This approach is appropriate for male dogs, male cats, and small female cats. The benefit of this approach is that it allows use of the rigid cystoscope to assess the urinary bladder and the proximal urethra, thus providing superior image quality compared with the flexible fiber optic cystoscope in male dogs and semirigid cystoscope in male cats. It allows access to the urinary tract in small female cats with urethras that do not accommodate the rigid cystoscope. In addition, it enables more rapid stone removal in male patients compared with laser lithotripsy and stone basketing via flexible cystourethroscopy.³⁵ A 1-cm skin incision is made over the apex of the urinary bladder. The urinary bladder is isolated and a stay suture is

placed into the apex of the bladder. The bladder is then gently retracted toward the body wall and the stay suture at the bladder's apex is tacked to the patient drape. Two stay sutures are then placed on each lateral aspect of the bladder and tacked to the patient drape, thus positioning the apex of the urinary bladder at the body wall. A stab incision is made into the apex of the urinary bladder and a screw trocar with diaphragm is inserted into the stab incision. The rigid cystourethroscope is passed through the trocar into the urinary bladder to assess the wall of the urinary bladder and the ureteral orifices, as well as the proximal urethra. The flexible cystourethroscope is then passed through the trocar and into the urethra to assess the urethral lumen. The penile urethra of male cats cannot be assessed with this technique because it is too small to accommodate the 7.5-Fr flexible cystourethroscope.³⁶

Percutaneous perineal approach for rigid retrograde cystourethroscopy in male dogs

This approach is appropriate for large male dogs when rigid retrograde cystourethroscopy is desirable; for example, when interventional endoscopic procedures are required (eg, cystoscopic laser ablation of ectopic ureters).⁵⁷ Although cystoscopic laser ablation of ectopic ureters has been described in male dogs using flexible cystourethroscopy, it is not possible to have a ureteral catheter in place to protect the lateral wall of the ectopic ureter when using this approach.⁴³ In addition, it is significantly more technically challenging to perform cystoscopic laser ablation of ectopic ureters using a flexible cystourethroscope than it is to perform the procedure with a rigid cystourethroscope. This difficulty is primarily caused by limitations in image quality resulting from many flexible cystouroscopes being fiber optic scopes rather than videoscopes, as well as from limitations in the maneuverability of flexible cystouroscopes. The patient is placed in dorsal recumbency and the perineal region and preputial region are clipped and aseptically prepared with surgical scrub. The prepuce is flushed using 0.05% chlorhexidine acetate solution. A sterile drape is placed over the patient and holes are made in the drape over the midischial region and preputial region. A red rubber catheter is passed into the urethra and is palpated in the urethra in the midischial region. It is gradually backed out of the urethra until the tip of the catheter can be felt just distal to the midischial urethra. Saline is introduced through the catheter to distend the urethra. The urethra is stabilized between the finger and thumb of the endoscopist's nondominant hand, and an 18-gauge intravenous catheter is inserted into the urethra through the skin, making sure to orient the catheter dorsally toward the urinary bladder. Once saline is seen entering the hub of the cannula, the cannula is removed and the catheter is left in place. A 0.89-mm (0.035-inch) hydrophilic guide wire (Weasel Wire, Infiniti Medical) is passed through the catheter and into the urethra, and is allowed to curl within the urinary bladder. The catheter is then removed over the wire, and serial fascial dilators are passed over the guide wire, starting at 4 to 5 Fr and ending at 16 Fr. A 16-Fr vascular access sheath with dilator is then passed into the urethra and the hydrophilic guide wire is removed. The dilator is removed, and the rigid cystoscope is then introduced into the urethra through the vascular access sheath. The proximal urethra and urinary bladder are then assessed as described for female dogs. Because the vascular access sheath is within the urethral lumen, assessment of the entire accessible portion of the urethra requires the sheath and cystoscope to be concurrently withdrawn from the urethra. Following the procedure, a standard urinary catheter is passed and attached to a closed collection system. An indwelling urinary catheter is kept in place for at least 24 hours after the procedure.

COMPLICATIONS AND MANAGEMENT
Mild Trauma of the Urethra and Urinary Bladder

Low-grade urethral and bladder mucosal injury is common during cystourethroscopy and typically does not require any form of treatment (Box 5). Hematuria may be present for 2 to 3 days after cystoscopy.²²

Stranguria, Pollakiuria, and Inappropriate Urination

Patients occasionally have lower urinary tract clinical signs of varying severity following cystourethroscopy.²² These signs typically persist for 2 to 5 days after cystoscopy. Lower urinary tract signs seem to be particularly common in females following lithotripsy. The authors typically preemptively treat these patients with tramadol (2–3 mg/kg orally every 6–8 hours) or oral buprenorphine (0.01 mg/kg orally every 6–8 hours). If the urethra appears hyperemic and inflamed at the end of a cystoscopy procedure, topical bupivacaine (2 mg/kg diluted 50:50 with saline) can be administered through the cystourethroscope as it is withdrawn. Steroidal or nonsteroidal antiinflammatories can be used to reduce urethritis in patients that have significant

Box 5 Complications associated with cystourethroscopy and their management	
Complications Associated with Cystourethroscopy	
Complication	Management Options
Mild trauma to urethral wall or urinary bladder wall	No treatment necessary (self-limiting)
Urethritis (resulting in stranguria, pollakiuria, or inappropriate urination)	1. Pain control: tramadol (2–3 mg/kg every 6–8 hours) or buprenorphine (0.01 mg/kg orally every 6–8 hours) 2. Topical analgesia (2 mg/kg diluted 50:50 with saline administered through cystoscope or urinary catheter into urethral lumen) 3. Antiinflammatory therapy (nonsteroidal antiinflammatory drug vs weaning course of prednisone) 4. Urethral antispasmodic therapy (prazosin 1 mg/15 kg orally every 8 hours in dogs; 0.5 mg total dose orally every 8–12 hours in cats)
Urinary incontinence	No treatment necessary (self-limiting)
Urethral perforation	Indwelling urinary catheter with closed collection system for 7–10 days
Bladder rupture	1. Indwelling urinary catheter with closed collection system for 7–10 days (only for small rents) 2. Surgical bladder wall closure
Urine passage through perineal access site (applies only to rigid cystoscopy in male dogs using the percutaneous perineal approach)	Indwelling urinary catheter for 24–72 hours postprocedure

clinical signs.²⁶ In addition, a urethral antispasmodic, such as prazosin (1 mg/15 kg every 8 hours in dogs; 0.25–0.5 mg [total dose] orally every 8–12 hours in cats) can be prescribed.

Incontinence

Following cystoscopy, continent female patients rarely have incontinence, which is potentially caused by excessive distension of the urethra.²² This condition is self-limiting and does not require treatment, but may persist for 3 to 5 days after cystoscopy.

Urethral Perforation

This is a rare complication of cystourethroscopy and is typically caused by the use of excessive force and operator inexperience.^{2,22,58} Perforation can also be caused by inadvertent damage to the urethral wall during laser lithotripsy or urethral biopsy acquisition (see **Fig. 31B**). In most cases, urethral perforations can be treated with the placement of an indwelling urinary catheter. Most sources recommend maintenance of the catheter for 7 to 10 days to allow complete healing of the tear.^{59–61} However, some sources suggest that shorter periods of urinary catheterization (24–48 hours) should suffice.^{22,60} Rarely, surgical correction of the urethral perforation may be required if the defect is very large or circumferential.^{1,59–61}

Bladder Rupture

This is a very rare complication of cystourethroscopy and is typically caused by overdistension of the bladder with saline during the procedure. It can also occur if the procedure is performed soon after a cystotomy (<7 days) and care is not taken to avoid overdistending the bladder. This complication may potentially be managed with an indwelling urinary catheter for a minimum of 7 days. However, if the rent is large, surgical closure is recommended.¹

Urine Passage Through Perineal Access Site

The owners of male patients for which the percutaneous perineal approach for rigid retrograde cystourethroscopy is used should be warned that urine may pass through both the penis and the perineal access site for a few days following the procedure.^{21,57} In addition, minor bleeding and occasional blood clot passage may occur through both sites. These complications typically resolve within 7 days of the procedure (usually by day 3).⁵⁷ They are worst immediately following the procedure and gradually resolve over the next several days. For this reason, an indwelling urinary catheter should be placed and maintained with a closed collection for at least 24 hours after the procedure in order to decompress the bladder and bypass the urethral access site.

POSTPROCEDURAL CARE

The patient is monitored carefully for complete anesthetic recovery, and is kept in the hospital until the patient is observed urinating with a subjectively good urine stream. Patients with significant stranguria or that are unable to form a urine stream should be kept in the hospital. The patient is provided with pain control, antispasmodic therapy, and/or antiinflammatory therapies as needed, and is closely monitored for urine retention. If these interventions do not improve the patient's urinary patterns, then an indwelling urinary catheter is placed and maintained with a closed collection system to allow the urethra to be bypassed for a brief period of time as the urethritis subsides.

Most patients are discharged from the hospital within a few hours of an uncomplicated diagnostic cystourethroscopy, but are usually hospitalized overnight following procedures such as laser lithotripsy or cystoscopic guided laser ablation of ectopic ureters. For pets with sterile urine before cystoscopy, some investigators prescribe a 3-day course of amoxicillin (10–20 mg/kg orally every 12 hours) to treat inadvertent urinary tract bacterial contamination during the procedure; however, this practice has not been critically evaluated.^{21,22,26,55}

OUTCOMES

The outcomes of cystoscopy depend on the goals set before performing the procedure. Cystoscopy can be purely a diagnostic procedure or it may be combined with therapeutic procedures. Cystoscopy can be used to diagnose problems that can then be treated using other modalities such as surgery or interventional radiology. For example, urethral strictures (**Fig. 32**) diagnosed via cystoscopy can be treated via urethral balloon dilation or urethral stent placement, both of which typically require fluoroscopic guidance. Patients with urethral obstruction caused by urethral masses may be treated with urethral stent placement (**Fig. 33**). Vaginourethral fistulae require surgical intervention to be corrected. In contrast, there are disease processes that can be treated via cystoscopic guided procedures. Examples include cystoscopic laser ablation of intramural ectopic ureters, removal of cystic or urethral calculi via lithotripsy, percutaneous cystolithotomy, cystoscopic guided laser resection of cystic polyps, laser ablation of vestibulovaginal septal remnants, extraction of remnant suture material, sclerotherapy for idiopathic renal hematuria, and injection of submucosal bulking agents or botulinum toxin for urethral sphincter mechanism incompetence.^{3,26,29–35,37,43,44,46,49–52}

Reporting, Follow-up, and Clinical Implications

A standardized cystoscopy reporting form is valuable to ensure that all anatomic structures are evaluated and abnormalities are recorded. The combination of a

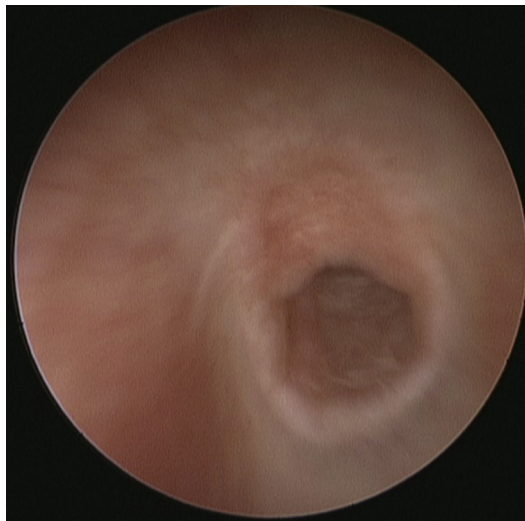


Fig. 32. Cystoscopic image from a female spayed dog of unknown age with a urethral stricture.

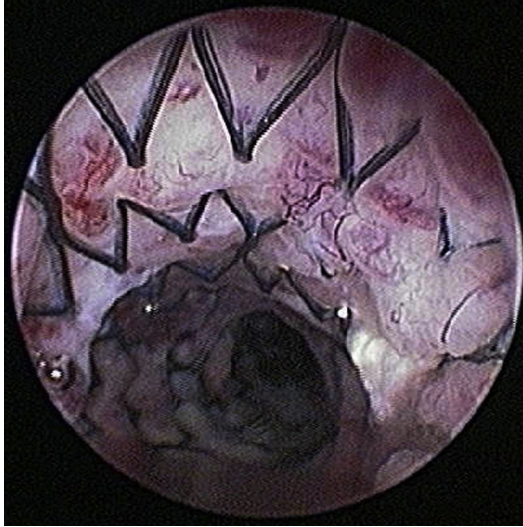


Fig. 33. Cystoscopic image from a female spayed dog of unknown age with a urethral transitional cell carcinoma after stent placement.

standardized cystoscopy report and images (still images and/or video) provides documentation for the medical record and facilitates communication with the owner and consulting veterinarians. Standardized reports have been published for female and male cystoscopy that also permit collection of data for prospective or retrospective research data, including grading urinary bladder mucosal abnormalities at low and high pressures and the volume of fluid to distend the urinary bladder to 80 cm of pressure. The authors recognize the benefits of these reports, but propose a simplified report (**Boxes 6 and 7**). Minimal follow-up is required immediately following diagnostic cystoscopy without therapeutic procedures. Because the procedure is performed under general anesthesia, standard postanesthesia recommendations are made for monitoring the pet until completely awake from anesthesia and for the reintroduction of food and water. Long-term follow-up depends on the cystoscopic diagnosis, results of histopathology, stone analysis results, culture results, and outcome following therapeutic interventions.²

CURRENT CONTROVERSIES/FUTURE CONSIDERATIONS

One of the limiting issues that have delayed the advancement of cystoscopy and urinary tract interventional endoscopy in dogs and cats is the lack of equipment that is of an appropriate size for these patients. There are gradual advancements being made in this regard, and the hope is that, as technology advances, the ability of veterinarians to treat urinary tract disease in a noninvasive fashion will progress. A current source of controversy in veterinary medicine is how best to treat urolithiasis. Advocates of traditional cystotomy cite its lower cost, more rapid procedure times, and universal availability as benefits.⁶² However, as is the case in human medicine, many veterinary clinicians are currently attempting to advance veterinary urology with the use of less invasive modalities that may reduce length of hospital stay and postprocedural complications. In addition, given that up to 20% of dogs treated with traditional cystotomy have incomplete urolith extraction, there is evidence that there is significant room to

Box 6**Female cystoscopy report**

Procedure length:

Anesthesia used:

Patient position

Dorsal []

Right lateral []

Sternal []

Miscellaneous:

Emergency procedure []

Fluoroscopy []

Computed tomography []

Procedural radiography []

Other []

Equipment used:

Cystoscope: 1.9-mm diameter, 18 cm long with a 10-Fr sheath []

Cystoscope: 2.7-mm diameter, 18 cm long with a 14.5-Fr sheath []

Cystoscope: 3.5-mm diameter, 30 cm long with a 17-Fr sheath []

Flexible biopsy forceps []

Stone basket []

Other []

Findings

Grades

Normal/absent = 0

If abnormal: mild = 1, moderate = 2, severe = 3; and describe

Urethra

Quality of examination:

External urethral meatus []

Vascularity []

Calculi/crystals []

Erosions []

Diverticulum []

Mass lesions []

Ectopic ureteral opening []

Urinary bladder

Quality of examination:

Left ureter opening []

Right ureter opening []

Trigone []

Vascularity []

Calculi/crystals []

Erosions []

Diverticulum []

Mass lesions []

Ectopic ureteral opening []

Vessel tortuosity []

Mucous/amorphous debris []

Hemorrhage []

Glomerulations []

Edema []

Granularity []

Inadequate insufflation []

Vagina

Quality of examination:

Frenula (remnants of paramesonephric septae) []

Vascularity []

Erosions []

Mass lesions []

Ectopic ureteral opening []

Mucous/amorphous debris []

Hemorrhage []

Granularity []

Therapeutic procedures:

Biopsies [], number taken []

Basket retrieval []

Lithotripsy []

Voiding urohydropulsion []

Stent placement []

Collagen implants []

Image storage:

Electronic []

Printed []

Complications:

During procedure []

Postprocedure []

Cystoscopic diagnosis:

Clinician performing procedure:

Box 7

Male cystoscopy report

Procedure length:

Anesthesia used:

Patient position

Right lateral []

Sternal []

Miscellaneous:

Percutaneous perineal approach []

Emergency procedure []

Fluoroscopy []

Computed tomography []

Procedural radiography []

Other []

Equipment used:

Cystoscope: 1.2 mm, flexible []

Cystourethroscope: 1.0 mm, semirigid []

Cystoscope: 7.5 Fr, 100 cm long, flexible []

Flexible biopsy forceps []

Stone basket []

Other []

Findings

Grades

Normal/absent = 0

If abnormal: mild = 1, moderate = 2, severe = 3; and describe

Urethra

Note location of abnormalities (preprostatic or postprostatic)

Quality of examination:

Quality of urine stream []

External urethral meatus []

Vascularity []

Calculi/crystals/plug []

Erosions []

Diverticulum []

Mass lesions []

Ectopic ureteral opening []

Prostatic openings []

Urinary bladder

Quality of examination:

Left ureter opening []

Right ureter opening []
 Trigone []
 Vascularity []
 Calculi/crystals []
 Erosions []
 Diverticulum []
 Mass lesions []
 Ectopic ureteral opening []
 Vessel tortuosity []
 Mucous/amorphous debris []
 Hemorrhage []
 Glomerulations []
 Edema []
 Granularity []
 Inadequate insufflation []

Therapeutic procedures:

Biopsies [], number taken []
 Basket retrieval []
 Lithotripsy []
 Voiding urohydropulsion []
 Stent placement []
 Collagen implants []

Image storage:

Electronic []
 Printed []

Complications:

During procedure []
 Postprocedure []

Cystoscopic diagnosis:

Clinician performing procedure:

improve the success of urolith extraction in veterinary medicine.⁶³ It is the opinion of the authors that the specifics of an individual case (such as stone size, number of stones, stone composition [if known], sex, and size of patient) should always dictate the procedure used for stone removal, and that cystotomy, percutaneous cystolithotomy/laparoscopic assisted cystoscopy, laser lithotripsy, and voiding urohydropulsion are all techniques that are appropriate under certain circumstances. Another common source of controversy in veterinary medicine is the best approach for the treatment of ureteral ectopia. There should be no argument that extramural ectopic ureters are best treated surgically. However, in the case of intramural ectopic ureters, particularly in female dogs, surgical correction is more costly and invasive than cystoscopic laser

ablation, and there is a higher postoperative complication rate.^{43,44,46} In addition, cystoscopic laser ablation offers the benefit of concurrent diagnosis and treatment of ureteral ectopia. Therefore, the authors support the use of cystoscopic guided laser ablation of intramural ectopic ureters in dogs.

SUMMARY

For experienced, well-trained clinicians, cystoscopy is a minimally invasive tool that has advanced urology in veterinary medicine. Cystoscopy provides the ability to diagnose conditions that cannot easily be diagnosed using other modalities. It also enables more thorough assessment and potentially treatment of conditions diagnosed using other modalities. As technology advances, a wider range of patients will be able to be assessed via cystoscopy, and more treatment modalities will become available. Advanced training and a wide range of specialized equipment are necessary to perform cystoscopy appropriately.

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